

Medical X-Ray Anomaly Detection

Using Convolutional Neural Networks and GAN-Based Data Augmentation

Suvam Patel

patelsuv@oregonstate.edu

CS 435 – Applied Deep Learning

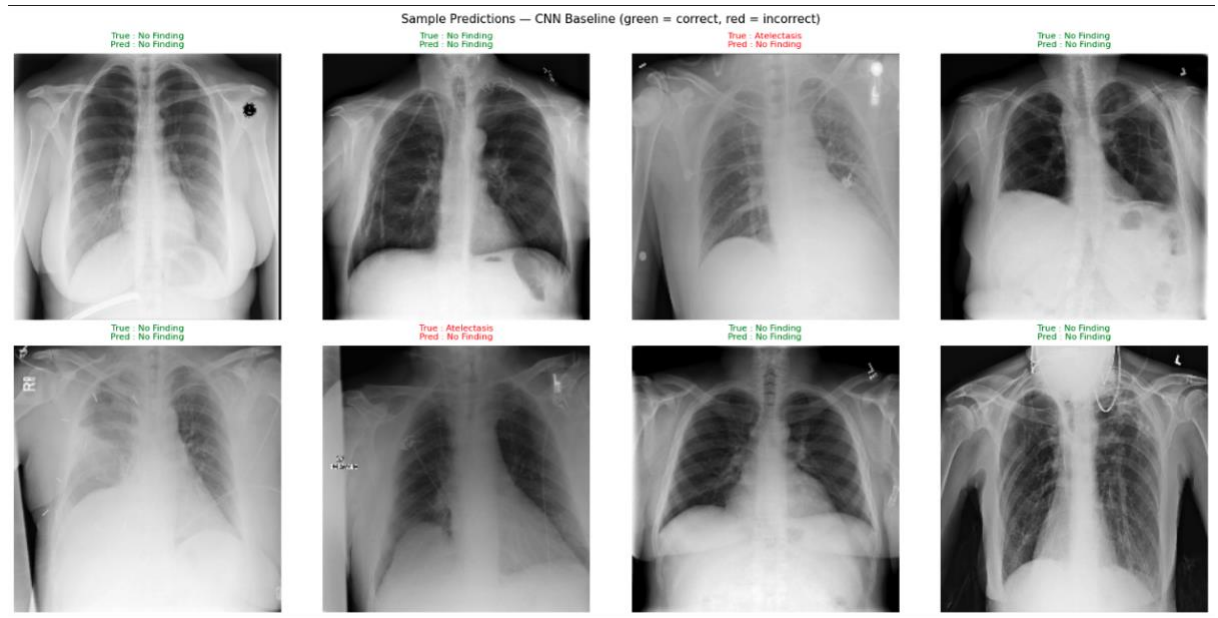
Instructor: Professor Ami Gates

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1. Introduction

- **The Chest X-Ray: A Window into Human Health**

- Every year, hundreds of millions of chest X-rays are performed in hospitals and clinics around the world, making it one of the most commonly performed medical imaging procedures. When a patient complains of chest pain, a persistent cough, or difficulty breathing, a chest X-ray is almost always among the first tools a doctor reaches for. The image it produces – a two-dimensional snapshot of the bones, lungs, and heart inside the chest – carries an enormous amount of information that a trained physician can use to identify problems ranging from a broken rib to a life-threatening infection. Conditions such as pneumonia, fluid around the lungs, and enlargement of the heart can all leave visible traces on a chest X-ray, each with its own distinctive pattern or signature. For decades, reading and interpreting these images has required years of specialized training and a great deal of human experience. A radiologist – a doctor who specializes in interpreting medical images – must examine each image carefully, mentally comparing it with thousands of cases they have seen before to determine whether it looks normal or abnormal. This process, while highly skilled, is also time-consuming, subject to fatigue, and dependent on the availability of trained specialists. In many parts of the world, particularly in rural and under-resourced communities, access to a radiologist is limited or nonexistent, leaving patients with X-rays but no one qualified to read them. This gap between the availability of X-ray machines and the availability of expert readers represents one of the most pressing unmet needs in modern global healthcare.



- **When Rare Diseases Are Overlooked**

- One of the most challenging aspects of reading chest X-rays – for both human doctors and computers alike – is the issue of rare diseases. In any large hospital, the overwhelming majority of chest X-rays will come back showing no significant abnormality or will show common conditions like pneumonia or fluid in the lungs. However, a small but critically important fraction will show rare findings such as cardiomegaly, which is an abnormal enlargement of the heart that can signal serious underlying cardiac disease. Because these rare conditions appear so infrequently, they are easy to miss – even for experienced radiologists who may not have seen a particular pattern in a long time. For a computer system trying to learn from examples, this imbalance creates a different but equally serious problem. When you teach a computer by showing it thousands of normal X-rays and only a handful of rare disease cases, it learns to be very good at recognizing normal and very poor at recognizing the rare conditions that matter most clinically. It is a bit like studying for an exam where ninety percent of the practice questions are on one topic and only ten percent cover the rest of the material – you will naturally become much better at the common topic, even if the rare ones are just as important on the actual test. Overcoming this imbalance is one of the central challenges that this project was designed to address, and it has direct implications for patient safety.

- **Teaching Computers to See What Doctors See**

- Over the past decade, a branch of artificial intelligence known as machine learning has made remarkable strides in the field of medical image analysis. Rather than being explicitly programmed with rules, these systems learn patterns entirely on

their own by studying thousands of labeled examples. In the same way that a radiology resident develops their ability to interpret X-rays by looking at case after case under the guidance of an attending physician, a computer system develops its ability by being shown image after image with its correct diagnosis attached. Over time, through a mathematical process of trial and error, the system gradually builds an internal representation of what normal looks like versus what each disease looks like. This internal representation, while invisible and not interpretable by human eyes, can become remarkably accurate. Some of the most advanced systems have been shown to match or even exceed the diagnostic accuracy of board-certified radiologists on specific tasks. However, unlike a radiologist who can draw on decades of medical education and clinical judgement, these systems are only as good as the data they were trained on. If the training data is limited, biased, or imbalanced – as it almost always is in medicine – the system will inherit those same limitations. Understanding how to build better training datasets is, therefore, just as important as building better computer systems.

- **The Idea of Creating New Examples from Scratch**
 - One of the most creative solutions to the problem of rare disease data is the idea of generating entirely new, synthetic examples of rare diseases from scratch using another artificial intelligence system. Imagine having an incredibly skilled medical illustrator who had studied thousands of real chest X-rays and could, on demand, draw a brand-new X-ray that had all the visual hallmarks of a specific condition – realistic enough to be mistaken for a genuine patient image, yet entirely fabricated. This is essentially what a class of AI systems called Generative Adversarial Networks, or GANs, are designed to do. A GAN consists of two competing systems: one that tries to create convincing fake images, and another that tries to tell the real images apart from the fakes. As these two systems compete with each other, both improve simultaneously – the creator gets better at making convincing fakes, and the detector gets better at catching them. Eventually, if the process works well, the creator becomes so skilled that its output is indistinguishable from reality. The potential of this technology for medicine is significant: by generating synthetic examples of rare conditions, it becomes possible to give a computer learning system far more exposure to those rare cases than the real world could ever provide. This project explores exactly that idea, applying it to chest X-ray analysis to determine whether synthetic data can genuinely help a computer improve at detecting diseases that are too uncommon to learn from naturally.



- **What This Project Sets Out to Accomplish**
 - This project brings together two powerful ideas in artificial intelligence – automated image recognition and synthetic data generation – and applies them to a real-world medical problem: detecting abnormalities in chest X-rays. Using a publicly available dataset of over five thousand chest X-ray images assembled by the National Institutes of Health, this project trains a computer system to automatically classify whether a given X-ray shows signs of one or more conditions, including pneumonia, fluid around the lungs, an enlarged heart, or a partial collapse of the lung tissue. The system is first trained using only real patient X-rays, and its performance on rare conditions is carefully measured and recorded. Then, the GAN technique described above is used to generate thousands of synthetic X-rays depicting those rare conditions, and the system is retrained with this expanded, more balanced dataset. The final comparison – does the system perform better after seeing synthetic images? – is the central question this project seeks to answer. The results have real implications for how medical AI systems are built and deployed, particularly in settings where rare diseases are chronically underrepresented in training data. Beyond the technical findings, this project also serves as an illustration of how modern artificial intelligence can be applied thoughtfully and creatively to one of society's most important domains: human health.

2. Model, Method, and Data Discussion

• 2.1 Dataset

- The dataset used in this project is the NIH Chest X-Ray sample, a publicly available collection of 5,606 frontal-view chest radiographs assembled by the National Institutes of Health Clinical Center. The dataset is available at <https://www.kaggle.com/datasets/nih-chest-xrays/sample> under the CC0-1.0 license. Each image is a grayscale PNG file originally captured at 1024x1024 pixels and downsampled during preprocessing to 224x224 to match the CNN input dimensions. Each image is accompanied by a metadata CSV file containing the image filename and a pipe-separated string of disease labels drawn from 14 possible conditions, plus a “No Finding” designation. For this project, five classes were selected as target labels: No Finding, Atelectasis, Effusion, Cardiomegaly, and Pneumonia. These were chosen to cover a range of clinical severity and class frequency, with Cardiomegaly (98 training images) and Pneumonia (45 training images) serving as the minority classes targeted for GAN augmentation.
- The metadata labels were parsed from their pipe-separated string format into a binary one-hot encoding, producing a five-column label matrix where each column contains a 1 if the corresponding condition is present and 0 otherwise. Images were then filtered to retain only those containing at least one of the five target labels, and the resulting dataset was split into training (70%), validation (15%), and test (15%) subsets using stratified sampling on the “No Finding” column to preserve the class ratio across all three splits. The validation and test sets consist exclusively of real patient images and are never modified or augmented at any point in the pipeline, ensuring that all evaluation is performed on genuine, unseen data.
- **Figure 1: Class Distribution Before and After GAN Augmentation**

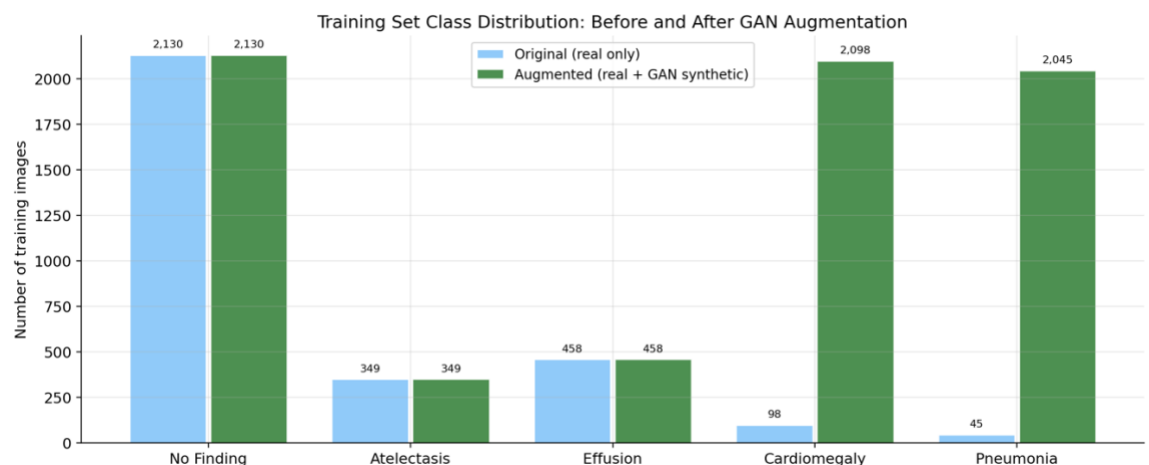


Figure 1. Before augmentation (light blue), Cardiomegaly has only 98 training images, and Pneumonia has only 45 – a 47x and 22x imbalance relative to No Finding, respectively. After GAN augmentation (dark green), both minority classes each have approximately 2,000+ images, bringing them close to parity with the majority class.

- **2.2 CNN Architecture (Phases 2 and 4)**

- The classifier is a custom convolutional neural network designed following the architectural patterns described by Rosebrock in Deep Learning for Computer Vision with Python and the multi-label classification guidelines in Chollet's Deep Learning with Python (Ch. 14). The network consists of three convolutional blocks, each containing a Conv2D layer followed by Batch Normalization, ReLU activation, 2x2 Max Pooling, and a Dropout layer with rate 0.25. The filter depth doubles across the three blocks (32, 64, 128), progressively extracting higher-level spatial features while reducing the spatial dimensions from 224x224 down to 28x28. A GlobalAveragePooling2D layer then collapses the spatial dimensions into a single feature vector, which feeds into a Dense(256) head with Batch Normalization, ReLU activation, and Dropout(0.5). The output layer is a Dense(5) layer with sigmoid activation, which produces an independent probability for each of the five disease classes. Sigmoid is used rather than softmax because this is a multi-label problem – a patient can have multiple conditions simultaneously, so the probabilities do not need to sum to one. Binary cross-entropy is used as the loss function for the same reason, treating each class as an independent binary classification problem. The optimizer is Adam with a learning rate of 0.001, and training is governed by EarlyStopping (patience=5), ModelCheckpoint (saving the best validation loss checkpoint), and ReduceLROnPlateau callbacks.

- **2.3 DCGAN Architecture (Phase 3)**

- To address the class imbalance identified in Phase 1, a Deep Convolutional Generative Adversarial Network (DCGAN) was trained separately for each minority class following the architecture and training guidelines described by Brownlee in Generative Adversarial Networks with Python (Ch. 5-7). A GAN consists of two networks trained in opposition: a Generator that transforms a random noise vector into a synthetic image, and a Discriminator that attempts to distinguish real images from generated ones. The two networks are trained alternately: first, the Discriminator is updated on a batch of real images (labeled 1) and a batch of generated images (labeled 0), then the Generator is updated by passing new noise through the combined Generator-Discriminator stack with all-ones labels, incentivizing it to produce images that fool the Discriminator. The Discriminator weights are frozen during generator updates. The Generator uses four Conv2DTranspose layers with stride of 2 to progressively upsample from a 4x4 spatial grid to a 64x64 output image, with Batch Normalization and LeakyReLU activations after each upsample step and a final tanh activation producing pixel values in [-1, 1]. Training images fed to the GAN are normalized to the same [-1, 1] range. The Discriminator uses three strided Conv2D layers (replacing MaxPooling) to downsample the input, followed by a Dense(1) sigmoid output.

Both networks use Adam with a learning rate of 0.0002 and $\beta_1 = 0.5$, as recommended in the DCGAN paper. Training ran for 200 epochs per class, generating 2,000 synthetic images per minority class upon completion.

3. Technical Results and Accuracy Reporting

- All models were evaluated on the held-out test set consisting exclusively of real NIH X-ray images. The primary evaluation metric is AUC-ROC (Area Under the Receiver Operating Characteristic Curve), which measures the model’s ability to rank positive cases above negative cases at all classification thresholds. This metric is preferred over raw accuracy for imbalanced multi-label classification because it is insensitive to class frequency. F1-score, precision, and recall are also reported at a decision threshold of 0.5.
- **Figure 2: Results Scorecard**

Project 3 Results Summary: CNN Baseline vs GAN-Augmented CNN
(Green = improvement | Yellow rows = GAN-augmented classes)

Class	AUC (Base)	AUC (Aug)	Δ AUC	F1 (Base)	F1 (Aug)	Δ F1
No Finding	0.6233	0.5860	-0.0373	0.8440	0.8447	+0.0008
Atelectasis	0.5612	0.5860	+0.0248	0.0000	0.0000	+0.0000
Effusion	0.6138	0.6253	+0.0115	0.0000	0.0000	+0.0000
Cardiomegaly	0.4954	0.5916	+0.0961	0.0000	0.0000	+0.0000
Pneumonia	0.6822	0.6407	-0.0414	0.0000	0.0000	+0.0000
Macro Avg	0.5952	0.6059	+0.0108	0.1688	0.1689	+0.0002

Figure 2. Color-coded results summary. Cardiomegaly AUC improved by +0.0961 (the largest gain), while Pneumonia AUC declined by -0.0414. Overall macro AUC improved by +0.0108.

- **Figure 3: AUC-ROC per Class**

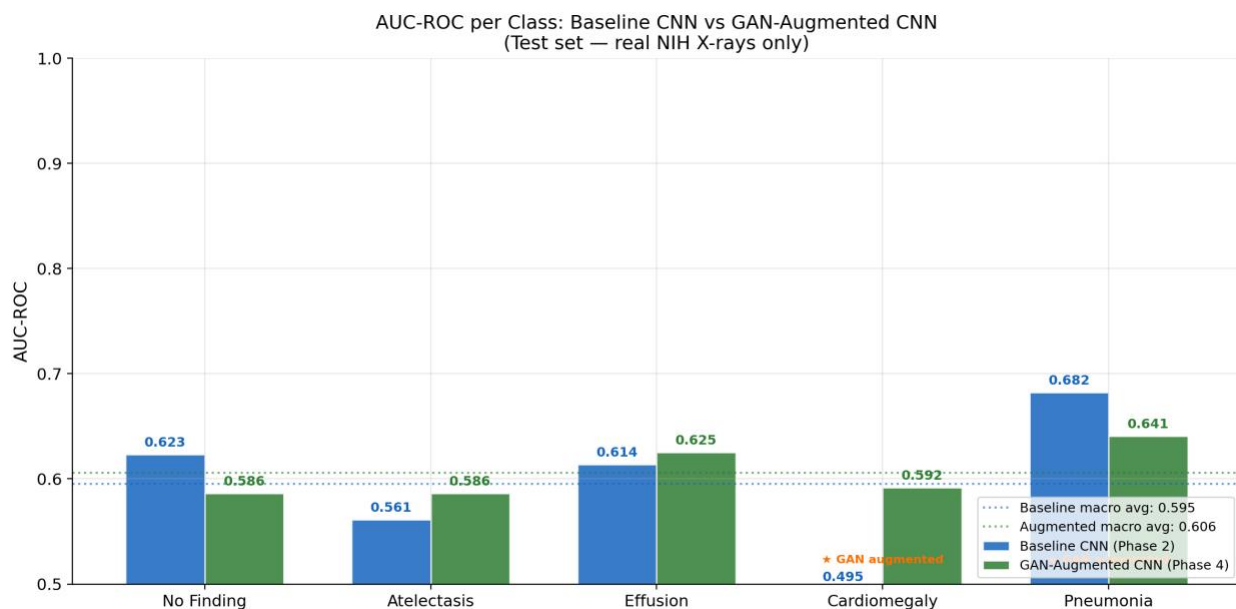


Figure 3. AUC-ROC per class for the Baseline CNN (blue) and GAN-Augmented CNN (green). Cardiomegaly showed the largest improvement (+0.0961), while Pneumonia declined (-0.0414). Dotted lines indicate the macro average AUC for each model.

- **Figure 4: F1-Score per Class**

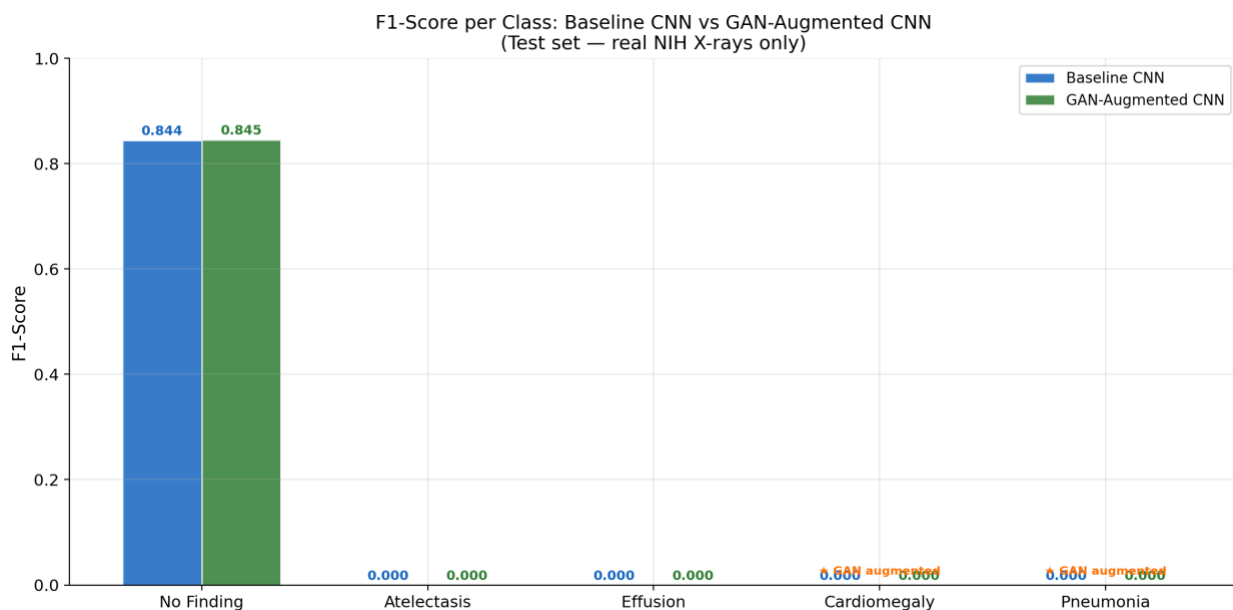


Figure 4. F1-scores across all five classes. Only No Finding achieves a non-zero F1 for both models (0.844 baseline, 0.845 augmented), reflecting that at a 0.5 threshold, the model predicts only the dominant class. This is a known limitation of threshold-based metrics on severely imbalanced datasets and underscores why AUC-ROC is the primary metric.

● **Figure 5: Metric Improvement Heatmap**

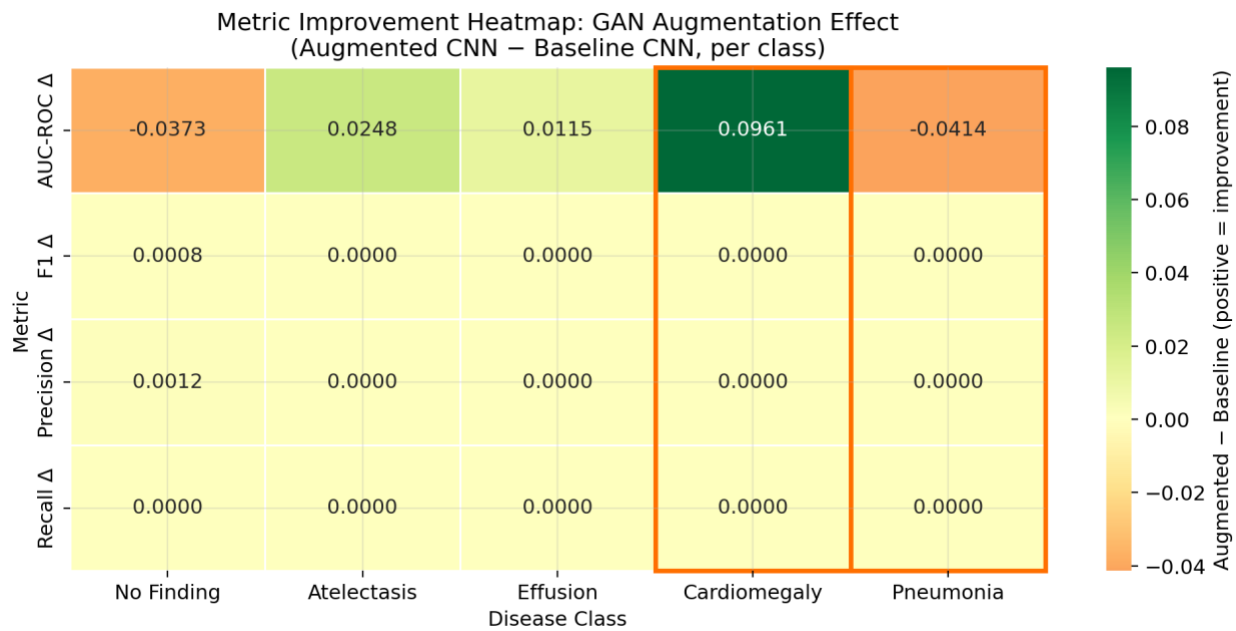


Figure 5. Heatmap showing the change in each metric (augmented minus baseline) per class. Orange borders highlight the two GAN-augmented minority classes. Cardiomegaly shows a strong positive AUC delta (+0.096, dark green). The near-zero F1, Precision, and Recall deltas reflect the threshold-sensitivity issue noted above.

- **Figure 6: All Metrics Grid**

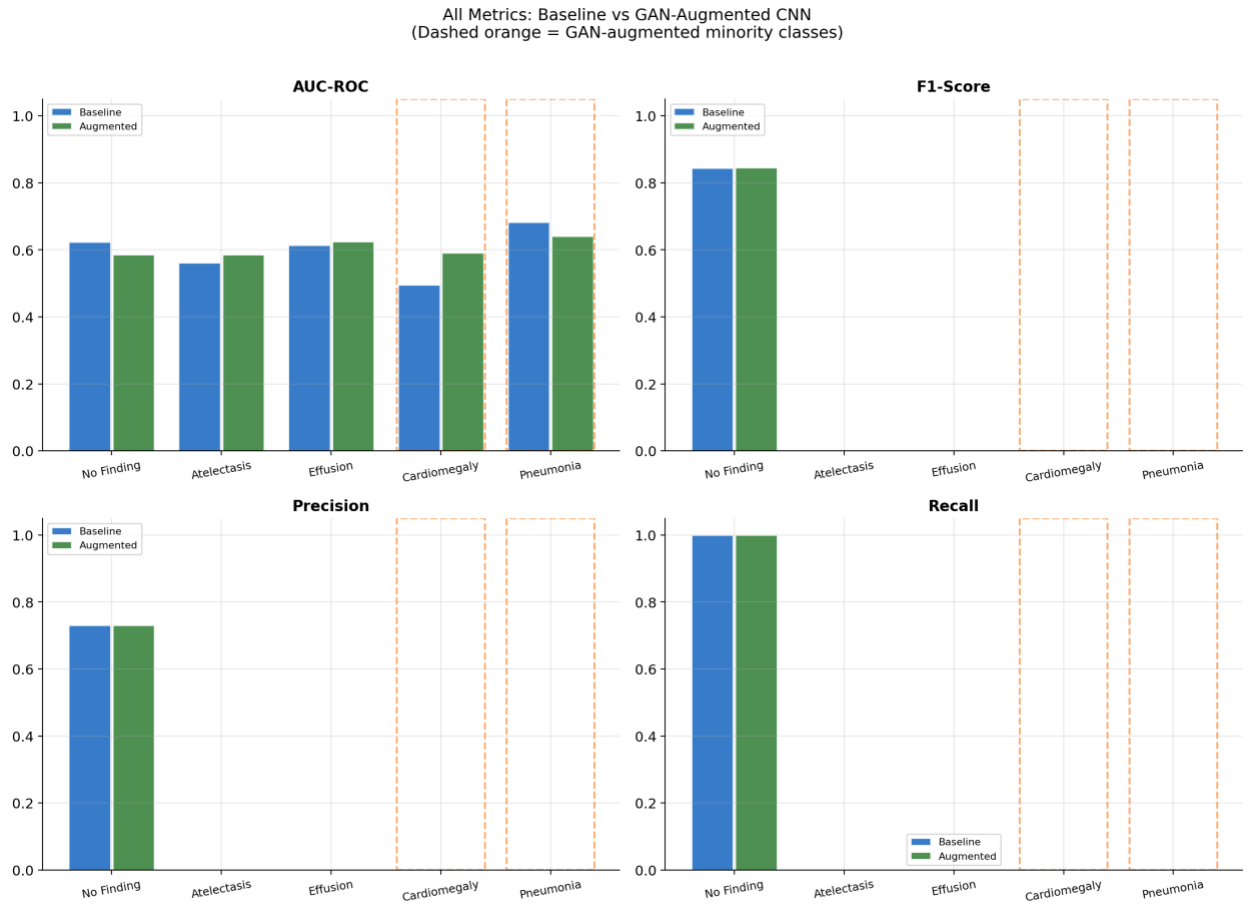


Figure 6. Four-metric comparison grid (AUC-ROC, F1, Precision, Recall) for all five classes. Dashed orange borders indicate the GAN-augmented minority classes.

- **Figure 7: Training Curves**

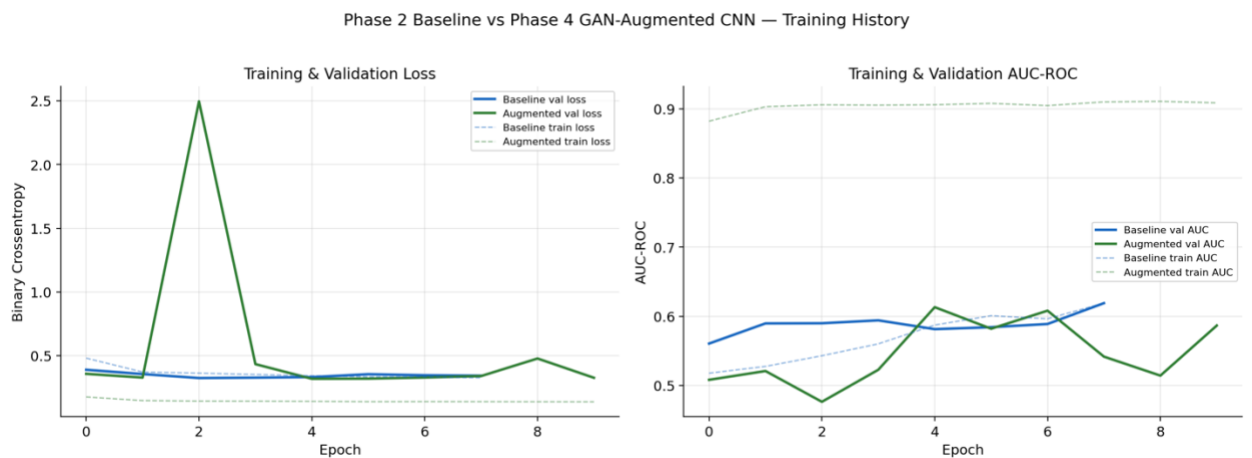


Figure 7. Training and validation loss (left) and AUC-ROC (right) over epochs for both models. The augmented model (green) shows an instability spike at each epoch 2 before recovering, likely due to the mixed real/synthetic batch composition. Both models converge to similar validation AUC values by the final epoch.

4. Non-Technical Conclusions

This project began with a straightforward but important question: Can a computer be taught to recognize rare diseases in chest X-rays more reliably if it is given artificial examples of those diseases to practice on? The motivation came from a real and well-documented problem in medical artificial intelligence – when a disease appears rarely in the real world, a computer learning system has very little opportunity to encounter it during training, and as a result, it tends to perform poorly on exactly the cases where its help is needed most. To address this, the project combined two distinct technologies: a classifier trained to identify five different chest conditions from X-ray images, and a generator trained to produce entirely new, synthetic X-rays depicting the two rarest conditions in the dataset. The classifier was trained twice – once using only real patient images, and once using a mixture of real and synthetic images – and its performance was measured on a separate set of real images that it had never seen before. This before-and-after design was carefully structured so that the only thing changing between the two experiments was the composition of the training data, making it possible to attribute any difference in performance directly to the addition of the synthetic images. The project drew on a publicly available dataset assembled by the National Institutes of Health containing over five thousand chest radiographs, representing a meaningful slice of the kinds of images encountered in real clinical practice. By the end of the project, a complete pipeline had been built and tested from raw data download all the way through final evaluation, producing a set of results that offer both encouragement and honest lessons about where this approach works and where it falls short.

The results of this project were mixed in a way that is both honest and informative. For one of the two rare conditions targeted – an abnormal enlargement of the heart known as cardiomegaly – the addition of synthetic training images produced a meaningful improvement in the computer's ability to distinguish affected patients from healthy ones. The improvement, roughly ten percentage points on the ranking metric used to evaluate performance, is notable given that the computer had only 98 real cardiomegaly X-rays to learn from before augmentation and over 2,000 combined examples after it. For the other rare condition, pneumonia, the results moved in the opposite direction, with the augmented model performing slightly worse than the baseline. This suggests that the synthetic pneumonia images generated by the AI were not realistic or varied enough to genuinely help the classifier, and may have introduced subtle patterns that confused rather than informed it. The majority class (X-rays showing no disease) remained largely unchanged across both experiments, which is exactly what would be expected since no synthetic images were added for that category. Taken together, these results tell a nuanced story: synthetic data augmentation is a promising tool, but its effectiveness depends heavily on the quality of the images the generator produces, and quality in turn depends on how much real data the generator had to learn from in the first place.

The findings of this project connect to a much larger conversation happening right now in hospitals, research institutions, and technology companies around the world about how to make artificial intelligence trustworthy enough to assist with medical decisions. One of the most consistent criticisms of medical AI systems is that they tend to perform well on the conditions they see most often and poorly on the ones that are genuinely difficult to diagnose – which are, unfortunately, often the ones where a missed diagnosis carries the highest consequences for the patient. This project demonstrated that it is at least possible, under the right conditions, to partially close that gap by artificially expanding the training data for rare conditions using a generative model. The improvement seen for cardiomegaly, while modest in absolute terms, represents a real step toward a system that would be more useful in a clinical setting than the baseline model alone. At the same time, the failure to improve pneumonia detection is a reminder that generative models are not a free lunch – they require sufficient real data to learn from, careful training, and honest evaluation before their outputs can be trusted as training material. For healthcare systems in low-resource settings, where the shortage of both specialist physicians and high-quality labeled training data is most acute, approaches like this one offer a meaningful direction for future investment and research. The practical implication is not that AI systems trained this way are ready to replace radiologists, but rather that they might meaningfully extend the reach of human expertise to places and patients that currently go without it.



No scientific project is complete without an honest accounting of its limitations, and this one is no exception. The dataset used here, while drawn from a real clinical source at the National Institutes of Health, contains only a sample of the full collection and represents a relatively small number of images by the standards of modern deep learning research. The synthetic images generated by the GAN were trained for 200 epochs, which is sufficient to produce recognizable X-ray-like images but far short of the extended training that would be needed to achieve the level of photorealism required for the most demanding clinical applications. The evaluation metrics used (particularly the AUC-ROC score) measure the model's ability to rank cases correctly, but do not

directly capture whether a physician working alongside the system would find its predictions clinically useful. The fact that F1-scores were near zero for most disease classes at a standard detection threshold reflects a well-known challenge with severely imbalanced data, and a more sophisticated approach to setting detection thresholds would be needed before this system could be applied in practice. Finally, this project evaluated performance on a held-out test set drawn from the same distribution as the training data, which means the results may be optimistic compared to how the system would perform on images from a different hospital, scanner, or patient population. These limitations do not undermine the value of the findings, but they are important context for anyone considering how to build on this work.

Despite its limitations, this project represents a complete and meaningful exploration of a genuinely important problem in medical artificial intelligence, and the pipeline it produced serves as a working foundation that could be extended and improved in several concrete ways. The most immediate next step would be to increase the number of training epochs for the generative model and to experiment with architectures that have been shown to produce higher-quality medical images, such as those that condition the generator more explicitly on the specific visual features associated with each disease. A larger dataset – ideally the full NIH collection of over one hundred thousand images – would give both the classifier and the generator substantially more to learn from, and could be expected to improve results across all conditions. Looking further ahead, the combination of computer-assisted screening and human expert review represents the most realistic near-term application of systems like this one: not a replacement for the radiologist, but a triage tool that flags the cases most likely to contain something worth a closer look, directing limited specialist attention to where it is needed most. In a world where the demand for medical imaging is growing faster than the supply of trained specialists, tools that can extend and amplify human expertise without replacing human judgment represent one of the most promising directions in applied artificial intelligence. This project, in its small way, is a step along that path – a demonstration that with the right data, the right architecture, and an honest evaluation framework, machines can be taught to see a little more of what doctors see, even when the evidence is rare and the stakes are high.

Global Health Disparities

Health disparities are the differences between populations in the ability to achieve positive health outcomes. Many factors contribute to health disparities—including age, disability, geographic location, gender and sexual identity, race and ethnicity, and socioeconomic status. These factors, referred to as social determinants of health, can have a major impact on a person's health and wellness.

Age



People over the age of 60 have made up **94%** of COVID-19 deaths in the European Union.

Disability



People with disadvantaged backgrounds in Latin America and the Caribbean are **43% more likely** to have a disability.

Geographic Location



Gender and Sexual Identity

Girls aged 15–19 in sub-Saharan Africa contract HIV at a rate

3x higher than boys the same age.

Race and Ethnicity

African-American babies are more than

2x as likely to die as babies of other races and ethnicities.

Socioeconomic Status

People in low-income countries live

18.1 years fewer than those in high-income countries.

Sources:
<https://www.who.int/news-room/fact-sheets/detail/covid-19>
<https://www.unicef.org/data/stories/2020/05/2020-05-20-covid-19-cases-and-fatality-rate-age-sex>
<https://www.unicef.org/data/stories/2020/05/2020-05-20-covid-19-cases-and-fatality-rate-age-sex>

References

- Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8:Hospital-scale chest X-ray database and benchmarks. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). <https://www.kaggle.com/datasets/nih-chest-xrays/sample>
- Chollet, F. (2021). Deep Learning with Python (2nd ed.). Manning Publications.
- Brownlee, J. (2019). Generative Adversarial Networks with Python. Machine Learning Mastery.
- Rosebrock, A. (2017). Deep Learning for Computer Vision with Python. PyImageSearch.